

Problem 2

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Problem 1 (a) *If a function is positive and decreasing on an interval $[a, b]$, will a right Riemann sum underestimate or overestimate the area of the region under the graph of the function? Justify your answer.*

Explanation. Because the function is decreasing, a right Riemann sum will be an underestimate. By definition, $f(x_k) < f(x)$ for all x in the interval $[x_{k-1}, x_k]$.

(b) *If a function is positive and decreasing on an interval $[a, b]$, will a left Riemann sum underestimate or overestimate the area of the region under the graph of the function? Justify your answer.*

Explanation. Because the function is decreasing, a right Riemann sum will be an overestimate. By definition, $f(x_{k-1}) > f(x)$ for all x in the interval $[x_{k-1}, x_k]$.